

EXPERIMENT – 5

AIM: To understand the operation of TELNET by accessing the router in a server room from a PC in IT Office.

THEORY:

TELNET

TELNET stands for Teletype Network. It is a type of protocol that enables one computer to connect to the local computer. It is used as a standard TCP/IP protocol for virtual terminal service which is provided by ISO. The computer which starts the connection is known as the local computer. The computer which accepts the connection is known as the remote computer. During telnet operation, whatever is being performed on the remote computer will be displayed by the local computer. Telnet operates on a client/server principle. The local computer uses a telnet client program, and the remote computers use a telnet server program.

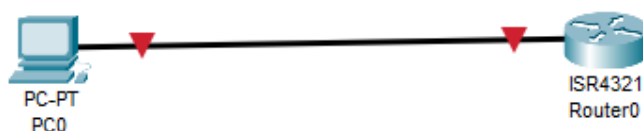
Types of Logging

- **Local Login:** Whenever a user logs into its local system, it is known as local login.
- **Remote Login:** Whenever a user logs into a remote computer and use services that are available on the remote computer.

Modes of Router

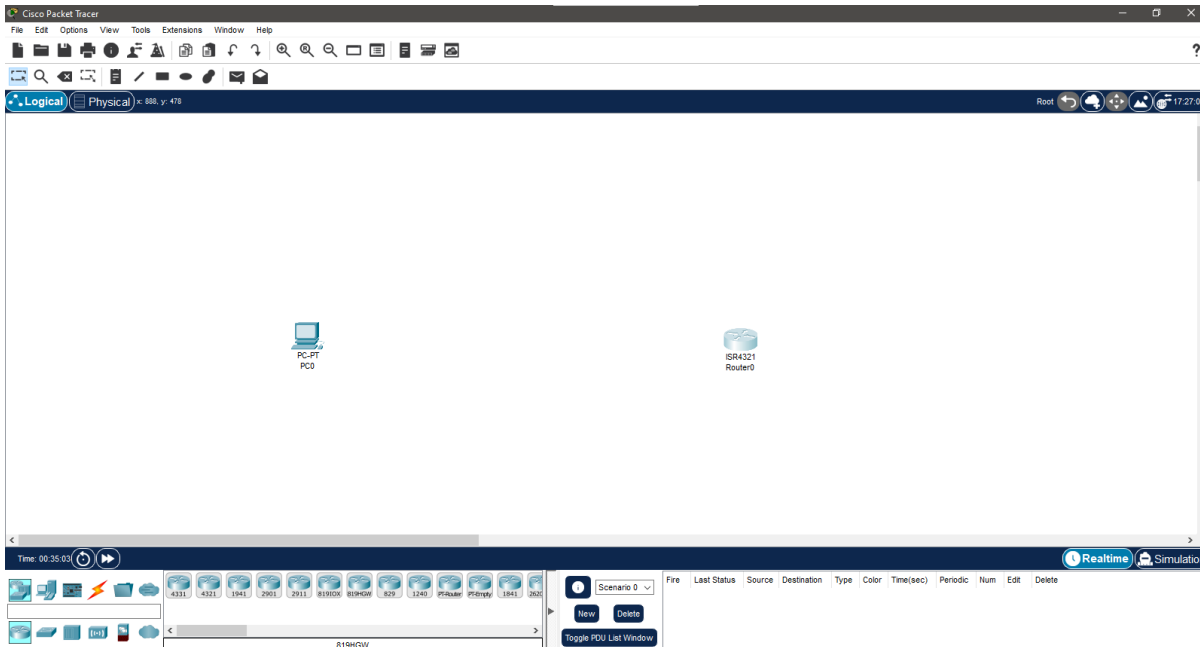
- **User execution mode:** As soon as the interface up message appears and press enter, the router prompt will pop up. This is called user execution mode. This mode is limited to some monitoring commands.
- **Privileged mode:** As we type enable to user mode, we enter into Privileged mode where we can view and change the configuration of the router.
- **Global configuration mode:** As we type configure terminal to the user mode, we will enter the global configuration mode. Commands entered in these modes are called global commands and they affect the running configuration of the router. In this mode, a different configuration like making a local database on the router by providing username and password can set enable and secret password, etc.
- **Interface configuration mode:** In this mode, only the configuration of interfaces is done Assigning an IP address to an interface, bringing up the interface are the common tasks done in this mode.

TOPOLOGY:

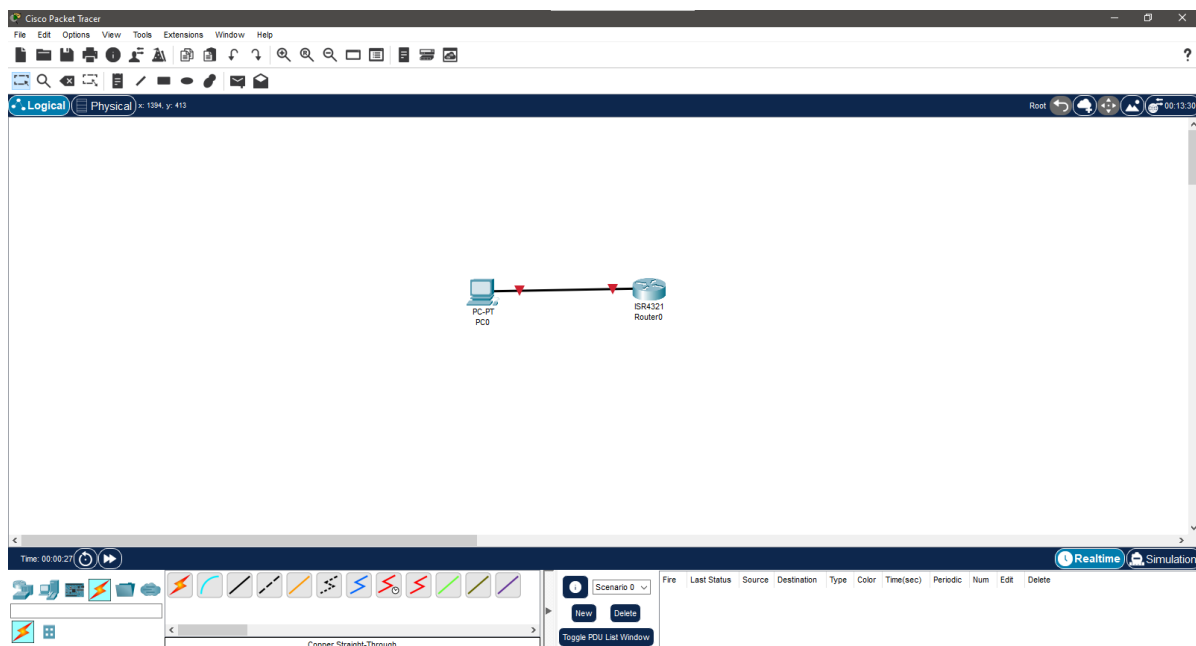


STEPS:

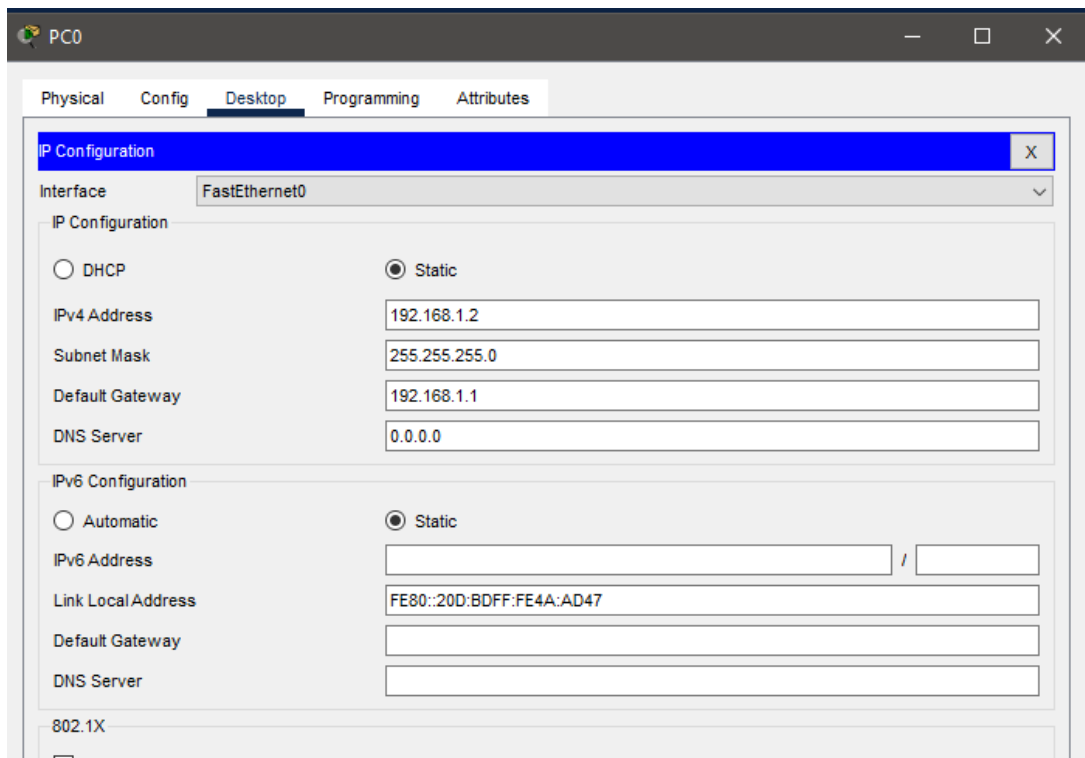
1. Open Cisco Packet Tracer and drag a PC and a 4321 router from Network devices category to the workspace.



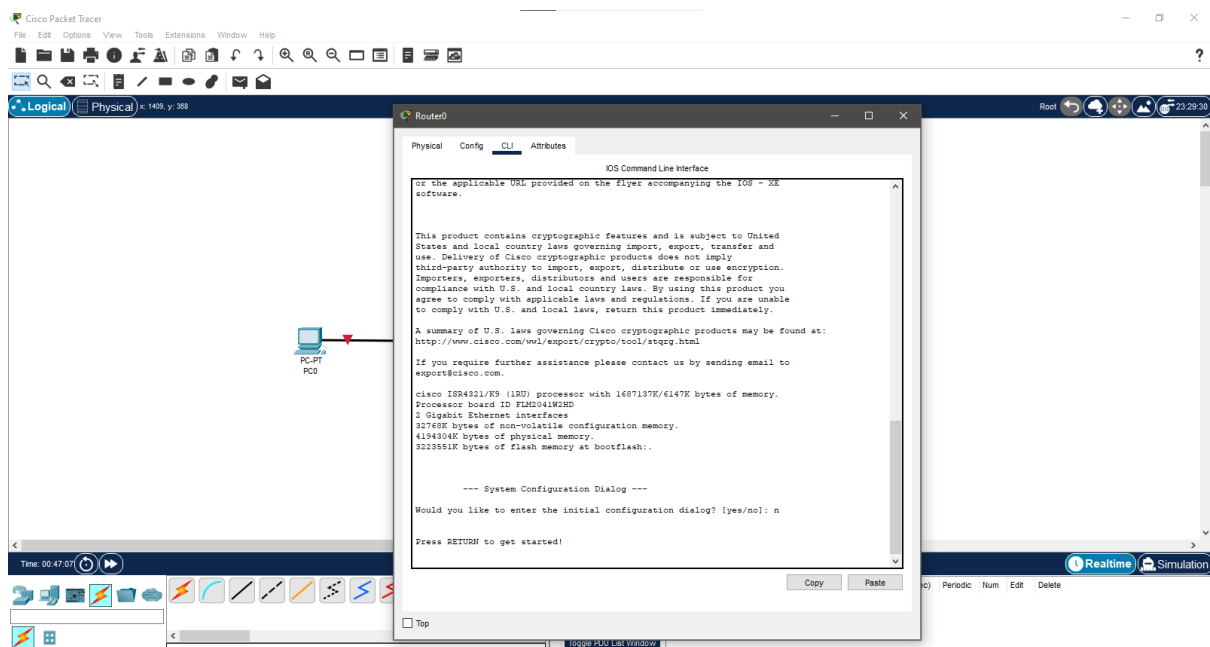
2. Connect the devices using Straight-Through cable with PC's FastEthernet() interface and Router's GigabitEthernet0/0/0 interface.



- Click on PC go to Desktop>IP Configuration and provide a static IP(eg: 192.168.1.1) and default gateway(eg: 192.168.1.2) to the PC.



- Click on router and go to CLI and enter N.

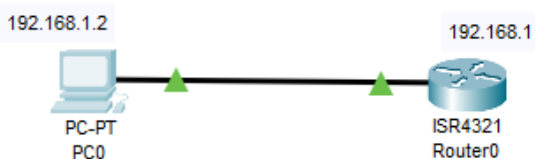


5. Enter the following bold marked commands in CLI:

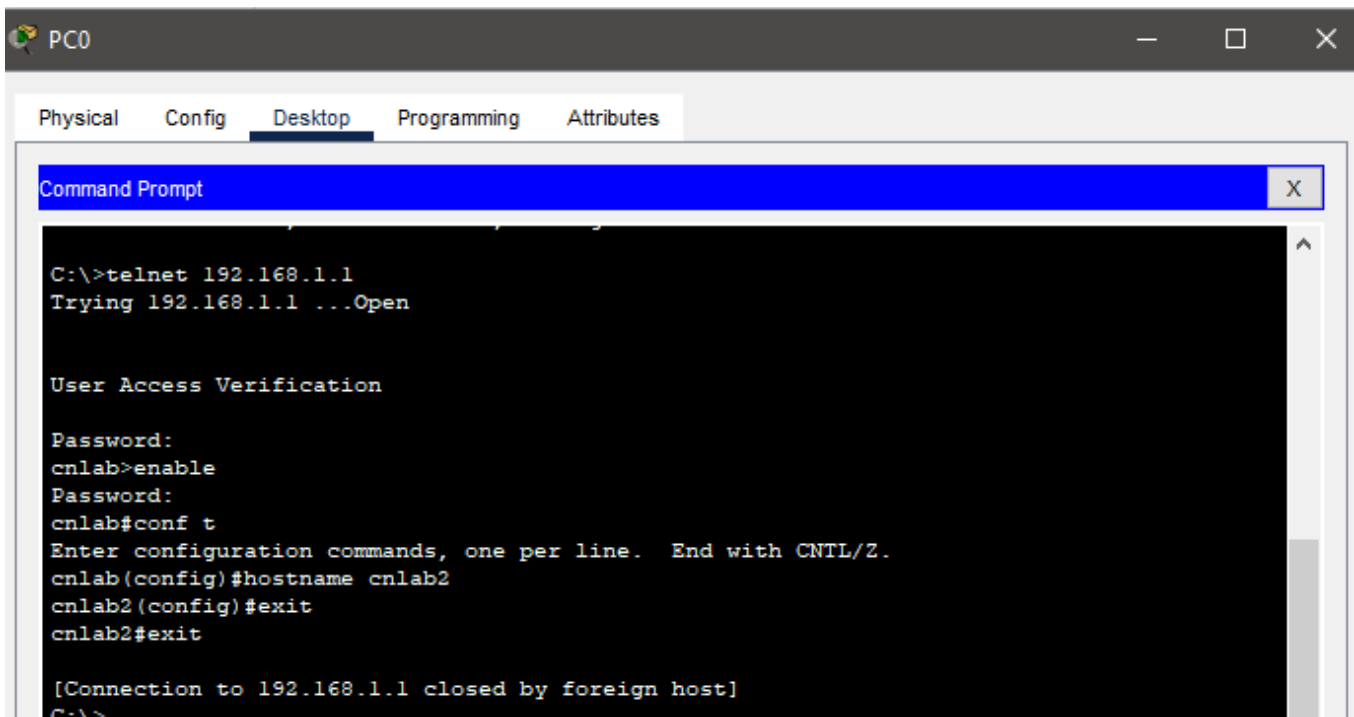
```
Router0
Physical Config CLI Attributes
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname cnlab
cnlab(config)#enable secret lab111
cnlab(config)#interface g0/0/0
cnlab(config-if)#ip address 192.168.1.1 255.255.255.0
cnlab(config-if)#no shut
cnlab(config-if)#
%LINK-S-CHANGED: Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
line vty 0 5
cnlab(config-line)#login
% Login disabled on line 2, until 'password' is set
% Login disabled on line 3, until 'password' is set
% Login disabled on line 4, until 'password' is set
% Login disabled on line 5, until 'password' is set
% Login disabled on line 6, until 'password' is set
% Login disabled on line 7, until 'password' is set
cnlab(config-line)#password abc
cnlab(config-line)#exit
cnlab(config)#exit
cnlab#
%SYS-5-CONFIG_I: Configured from console by console
```

6. Ping to verify the connection

```
PC0
Physical Config Desktop Programming Attributes
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```



7. Use TELNET to change hostname.



```
C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open

User Access Verification

Password:
cnlab>enable
Password:
cnlab#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
cnlab(config)#hostname cnlab2
cnlab2(config)#exit
cnlab2#exit

[Connection to 192.168.1.1 closed by foreign host]
C:\>
```

RESULT

Name changed from cnlab to cnlab2.



```
Router0

Physical  Config  CLI  Attributes

IOS Command Line Interface

cnlab2>
```